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RAJAGIRI COLLEGE OF SOCIAL SCIENCES
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KALAMASSERY-KOCHI-683104**



**MASTER OF COMPUTER APPLICATIONS
2024 - 2026**

**INTERNSHIP REPORT
MCA405**

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INTERNSHIP REPORT



SEMESTER IV MASTER OF COMPUTER APPLICATIONS (2024-2026)

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MARCH 2026



DEPARTMENT OF COMPUTER SCIENCE
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CERTIFICATE

*This is to certify that the internship titled “Python Full-Stack Developer Trainee” is a bona fide work carried out by **RIYA MATHEW** register number **24204049** in partial fulfillment of the requirements for the award of the Master’s degree in Computer Applications of Rajagiri College of Social Sciences (Autonomous), affiliated to Mahatma Gandhi University, during the year 2024- 2026. This internship report has been approved as it satisfies the academic requirement of internship work prescribed for the Master of Computer Applications.*

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CERTIFICATE

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This certificate awarded to

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12th March, 2026



Krishnan G
Chief Executive Officer
GDK Techspot LLP

DECLARATION

I hereby declare that this Internship work and the report submitted to the Department of Computer Science, Rajagiri College of Social Sciences (Autonomous), Kalamassery in partial fulfillment of the award of the degree of Master of Computer Applications is an outcome of my own work.

To the best of my knowledge, this Internship work or parts there does not form a part of any other project work or thesis on the basis of which a degree or award was conferred on an earlier occasion.

Date :

Place : Kalamassery

Riya Mathew

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Riya Mathew

ABSTRACT

This internship report presents the work carried out and the learning outcomes achieved during the internship at GDK TechySpot LLP, an IT and EdTech startup organization. The internship commenced on 1st January and is planned for a duration of six to twelve months, providing extended exposure to real-world software development practices. The internship role undertaken was Python Full-Stack Developer Trainee, with responsibilities spanning full-stack application development, UI/UX design, testing, and deployment activities.

The primary objective of the internship was to gain practical exposure to modern software development practices, strengthen foundational programming knowledge and develop skills in workflow automation and artificial intelligence integration within web applications. The internship involved studying existing applications, developing new application modules and understanding frontend-backend interactions within real-world systems.

The internship began with a training period focused on strengthening knowledge in technologies such as Linux, HTML, CSS, JavaScript and Java. These concepts were practiced through exercises and implemented in a demo healthcare project titled Medcore Systems. During this stage, several frontend components were designed and developed. Interactive features were implemented to enhance usability and user interaction.

Following the training phase, project-based tasks were assigned involving workflow automation and AI integration. An AI-based social media automation workflow was developed using the n8n platform to generate summaries from blog URLs, create AI-generated images, and automate posting to social media platforms with an approval mechanism. In addition, AI workflows were integrated with a Django-based web application to support social media plan generation, automated posting, YouTube video downloading, and template-based image generation using extracted visual features.

Overall, the internship has provided valuable practical experience in web development, automation workflows, and AI integration within modern applications, helping bridge the gap between academic learning and industry practices.

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CHAPTER 1 - INTRODUCTION

1.1 ORGANIZATION PROFILE

GDK TechySpot LLP is a software and technology consulting organization that delivers scalable, intelligent, and future-ready digital solutions across multiple domains, including education, enterprise systems, and emerging technologies. The company focuses on designing, developing, and deploying software platforms that support digital transformation, data-driven decision-making, and efficient system management for organizations and institutions.

The organization's core areas of expertise include software development, web and application solutions, cloud-based platforms, data analytics, and integration of modern digital technologies. TechySpot collaborates with enterprises and innovation ecosystems to create technology-driven solutions aligned with industry standards and evolving digital requirements. By adopting structured development practices and modern technology stacks, the company aims to deliver reliable, scalable, and user-centric software products.

TechySpot operates with a vision of enabling technology-driven innovation and sustainable digital growth. Through its solution-oriented approach and focus on quality engineering practices, the organization delivers robust technological systems that support evolving business and institutional requirements.

1.2 WORK 1 – MEDICORE SYSTEMS DEMO PROJECT

1.2.1 BACKGROUND INFORMATION

Modern web applications require a combination of frontend technologies, scripting languages and structured programming concepts to create interactive and user-friendly systems. Understanding the fundamentals of web development and programming is essential before working on larger application architectures or automation systems. During the initial phase of the internship, emphasis was placed on strengthening foundational knowledge in technologies such as Linux, HTML, CSS,

JavaScript and Java which are important for developing interactive web applications and managing development environments.

In addition to theoretical learning, these technologies were practiced through exercises and tasks to better understand their practical applications. A demo project named Medcore Systems was also used as a practical platform to apply the learned concepts. Working on this project helped in understanding how different technologies interact within a complete application and how various components are developed and integrated to build a functional system.

1.2.2 INTRODUCTION TO THE WORK

As part of the internship training activities, development tasks were carried out on the demo healthcare project Medcore Systems to gain practical experience in frontend development by implementing user interface components and interactive features using web technologies. The work involved creating and modifying several pages of the application, including designing the homepage layout, implementing JavaScript-based functionalities to improve user interaction and developing login and registration pages to simulate basic user authentication interfaces while ensuring proper structure, visual consistency and responsiveness across different screen sizes.

Alongside the project work, time was dedicated to strengthening programming knowledge by revisiting Java fundamentals, understanding the compilation process and practicing object-oriented programming concepts and custom exception handling. This helped build a strong foundation in both web development and core programming concepts while gaining practical experience in a project environment.

1.2.3 SCOPE OF THE WORK

- Strengthening foundational development skills through practical implementation and structured learning.
- Designing and implementing frontend components for the Medcore Systems demo project.
- Developing interactive features using JavaScript to enhance user interaction.

- Designing forms for user input with basic client-side validation.

1.2.4 OBJECTIVES

- To gain practical experience in frontend web development through implementation in the Medico Systems demo project.
- To design and develop key webpage components including the homepage, login page, registration page and support page with interactive features.
- To implement dynamic interface functionalities such as FAQ toggles and client-side form validation using JavaScript.
- To strengthen understanding of core programming concepts and object-oriented principles through structured learning and practice.

1.3 WORK 2 – AI-BASED SOCIAL MEDIA AUTOMATION USING n8n

1.3.1 BACKGROUND INFORMATION

Social media platforms have become an important medium for marketing and communication for businesses and organizations. However, managing multiple social media accounts and posting content regularly requires significant manual effort.

Automation tools such as n8n allow developers to create automated workflows that integrate different services and perform tasks automatically. By combining automation platforms with artificial intelligence models, content creation and publishing processes can be automated efficiently.

1.3.2 INTRODUCTION

The objective of this work was to develop an automation workflow capable of generating social media content and publishing it automatically. The system uses AI models to generate summaries from blog URLs, create images based on the generated content, and post them to social media platforms such as Instagram and Facebook.

The workflow also includes an approval mechanism that allows users to review generated posts before they are published.

1.3.3 SCOPE OF WORK

The scope of the project included:

- Designing automation workflows using the n8n platform
- Integrating AI models for content summarization
- Generating images using AI image generation models
- Publishing automated posts on social media platforms
- Implementing approval mechanisms for generated posts
- Integrating cloud storage for image hosting

1.3.4 OBJECTIVES

- To learn workflow automation using the n8n platform
- To implement automated content generation using AI models
- To automate posting of content to social media platforms

1.4 WORK 3 - AI WORKFLOWS INTEGRATION WITH DJANGO APPLICATION

1.4.1 BACKGROUND INFORMATION

With the increasing adoption of artificial intelligence in web-based systems, many modern applications are integrating AI-driven workflows to automate tasks and improve user productivity. Web frameworks such as Django provide a robust backend environment that can be combined with automation platforms and AI models to build intelligent systems capable of performing complex operations with minimal human intervention. Integrating AI workflows within web applications allows users to access advanced functionalities such as automated content generation, information retrieval, and media processing through simple interfaces.

1.4.2 INTRODUCTION

This work involved integrating AI workflows with a Django-based web application. The system allowed users to generate social media plans, retrieve relevant information using Vision-Language Model, and automate content generation tasks. Additional functionalities such as image generation, video downloading, and data

extraction using AI models were implemented.

1.4.3 SCOPE OF WORK

The work involved:

- Automation of social media posting tasks
- Development of web interfaces for interacting with AI workflows
- Integration of Django web application with automation workflows

1.4.4 OBJECTIVES

- To integrate AI workflows with a Django web application
- To develop user interfaces for interacting with AI systems
- To automate social media content generation and publishing
- To enhance system efficiency through workflow optimization

CHAPTER 2 – METHODS AND DISCUSSIONS

2.1 WORK 1 – MEDICORE SYSTEMS DEMO PROJECT

2.1.1 METHODOLOGY

- Study of foundational technologies

Strengthened knowledge in HTML, CSS, JavaScript, jQuery, and Java programming. Reviewed concepts of object-oriented programming including encapsulation, inheritance, polymorphism, interfaces, abstract classes, collections, and custom exception handling.

- Practical application on a demo project

The Medicare Systems demo project was used to apply learned concepts in a real-world scenario. The tasks completed included creating and modifying web pages, designing user interfaces and implementing interactive functionalities.

- Frontend development implementation

Designed and developed the homepage, login, registration, and support pages. Dynamic elements such as FAQ toggles and form validation using JavaScript were added.

2.1.2 DESIGN AND DEVELOPMENT

The key technologies used:

- HTML and CSS – For structuring and styling web pages, ensuring a visually consistent and responsive layout.
- JavaScript – To implement interactive and dynamic features such as FAQ toggles and client-side form validation.
- VS Code – Development environments used for coding, editing, and running the project.

The key features developed include:

- **Homepage Design:** Developed a clean and organized layout with clear navigation and consistent styling for a user-friendly interface.
- **Login and Registration Pages:** Created structured forms with proper alignment, input fields, and validation logic for realistic user authentication simulation.
- **Support Page:** Implemented an interactive FAQ section allowing users to expand and collapse answers dynamically using JavaScript. Developed a **Contact Support form** with real-time input checks to ensure proper data entry.

2.1.3 KEY FINDINGS AND RESULTS

The development of the Medcore Systems demo project provided experience in frontend web development and the practical application of programming concepts. A functional and visually consistent homepage was created, offering clear navigation and an organized layout that improves user experience as shown in figure 2.1.1.

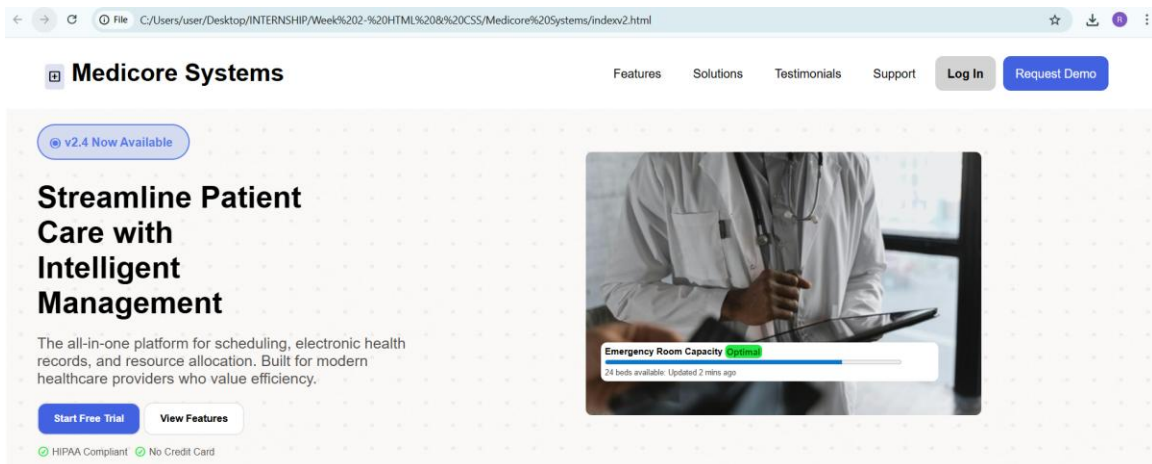
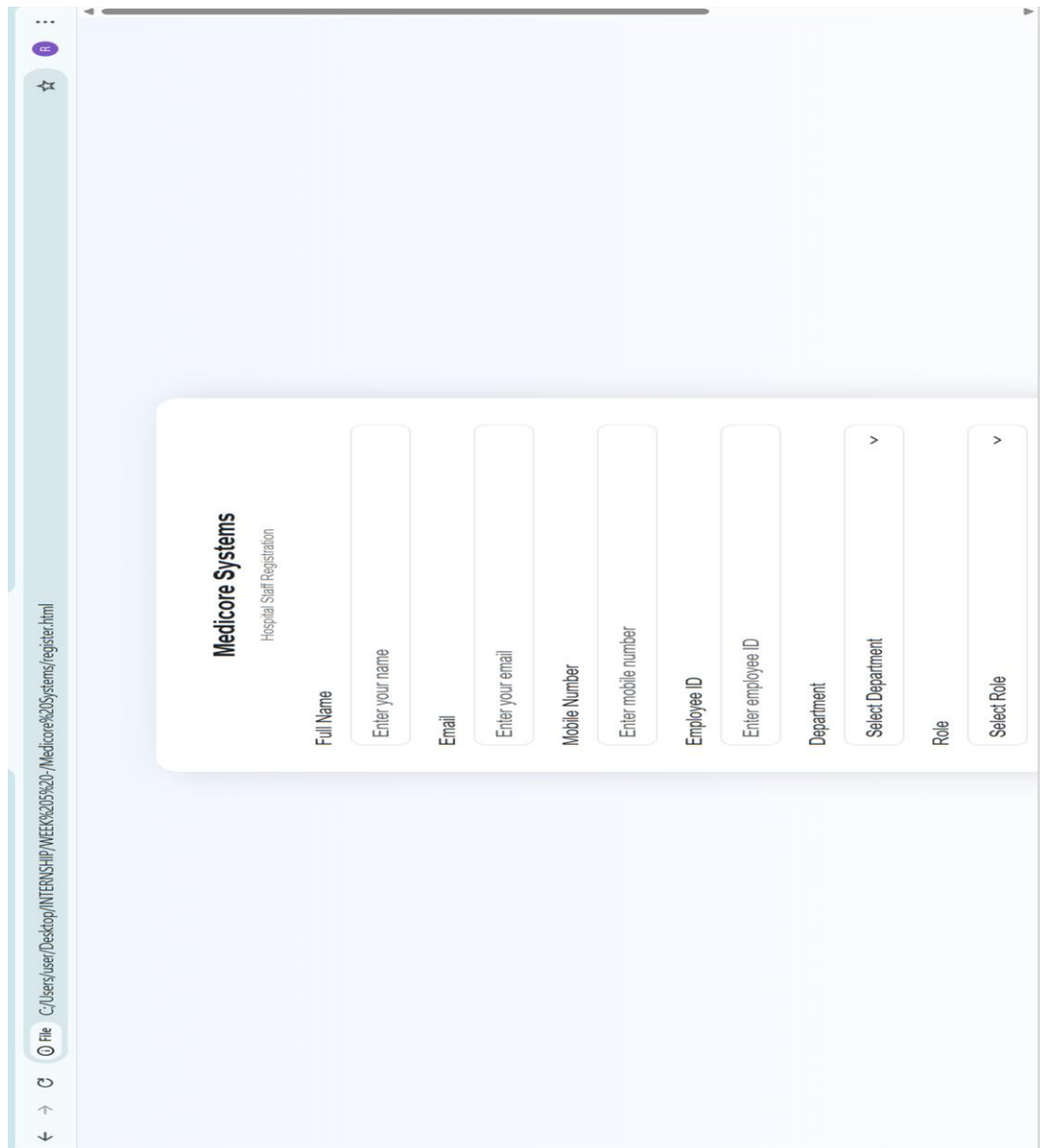


Figure 2.1. 1: Home page of Medcore Systems

Figure 2.1.2 shows the registration page. The login and registration pages were implemented with structured input fields, simulating real-world user authentication processes. This phase helped in understanding how different frontend technologies interact to create cohesive and user-friendly web interfaces.



The screenshot displays a web browser window with the address bar showing the file path: `C:\Users\user\Desktop\INTERNSHIP\WEEK9\2020-Medicore\20Systems\register.html`. The page content is a registration form titled "Medicore Systems" with the subtitle "Hospital Staff Registration". The form includes the following fields:

- Full Name**: A text input field with the placeholder "Enter your name".
- Email**: A text input field with the placeholder "Enter your email".
- Mobile Number**: A text input field with the placeholder "Enter mobile number".
- Employee ID**: A text input field with the placeholder "Enter employee ID".
- Department**: A dropdown menu with the placeholder "Select Department".
- Role**: A dropdown menu with the placeholder "Select Role".

Figure 2.1. 2: Registration page in Medicore Systems

The support page, as shown in Figure 2.1.3, demonstrated dynamic features through an interactive FAQ section, allowing users to expand and collapse questions seamlessly, and a Contact Support form that ensured proper input formatting in real time using JavaScript. By integrating HTML, CSS and JavaScript, the pages were made responsive. Overall, the project showcased how core web technologies can be used to build interactive, functional and user-friendly web applications.

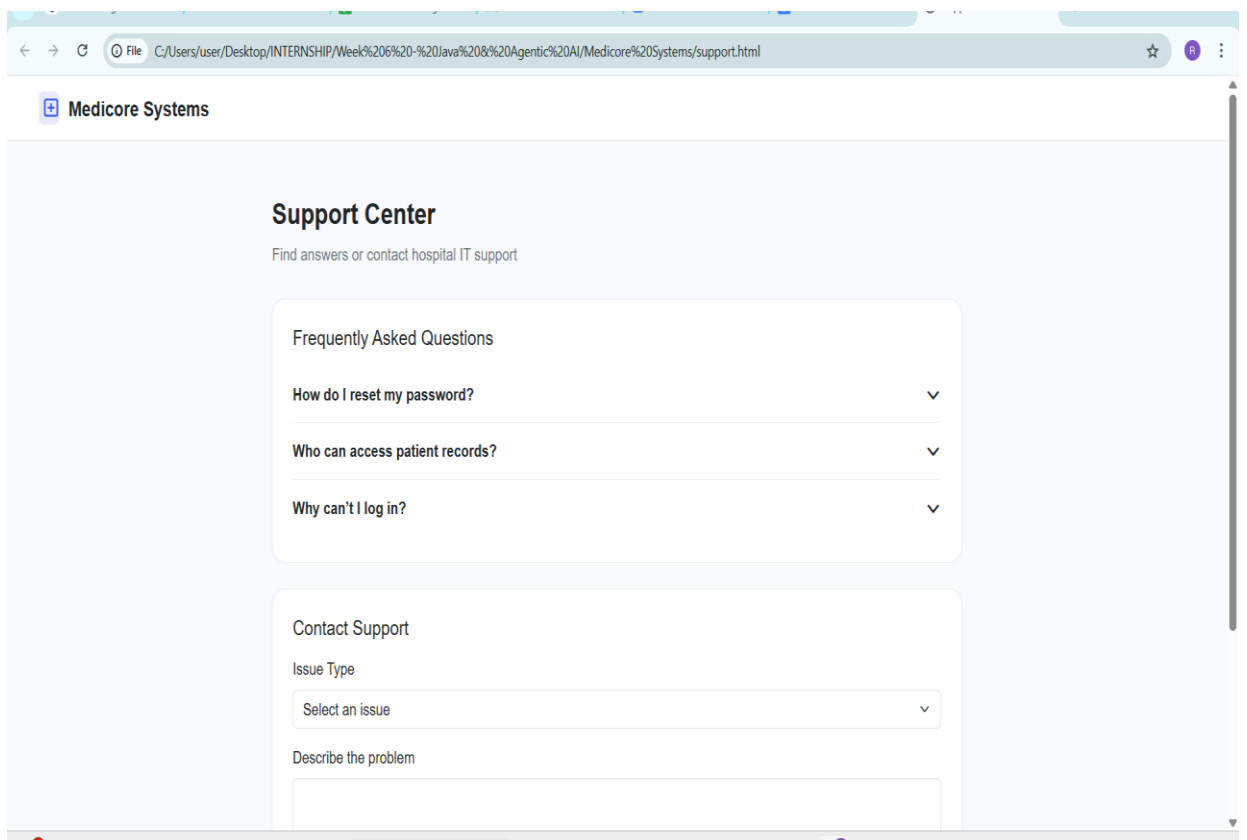


Figure 2.1. 3: Support Page

2.1.4 OBJECTIVES ACHIEVED

- Gained practical experience in frontend web development using HTML, CSS, JavaScript and jQuery.
- Strengthened understanding of core programming concepts and object-oriented principles through coding exercises.
- Designed and developed core web pages including homepage, login, registration and support pages.
- Implemented dynamic UI features such as FAQ toggles and client-side form validation using JavaScript.
- Developed the ability to build responsive and user-friendly interfaces while understanding real-world web application structure.

2.2 WORK 2 – AI-BASED SOCIAL MEDIA AUTOMATION USING n8n

2.2.1 METHODOLOGY

The workflow automation system was developed using the n8n platform. The system follows a series of steps to generate and publish content automatically.

- A blog URL is provided as input.
- The content of the blog is extracted.
- An AI model generates a summary of the blog content.
- An image is generated based on the summarized content.
- The generated image is stored in cloud storage.
- The generated post is sent for approval.
- After approval, the post is published to social media platforms.

2.2.2 DESIGN AND DEVELOPMENT

The tools used are:

- N8n: Workflow automation
- Groq AI: Text summarization
- Google Gemini: Image generation
- HuggingFace Models: used for alternate image generation

- Cloudinary: Image hosting so as to get public URL
- Docker: Containerization

2.2.3 KEY FINDINGS AND RESULTS

The implementation of the AI-based social media automation workflow using the n8n platform demonstrated that automation can significantly reduce the manual effort required for content creation and publishing. The workflow successfully processed blog URLs, extracted relevant content, generated summaries using Groq AI, and created images based on the summarized content. These summaries served as the primary text for social media posts while maintaining relevance to the original blog content. The workflow is shown in figure 2.2.1.

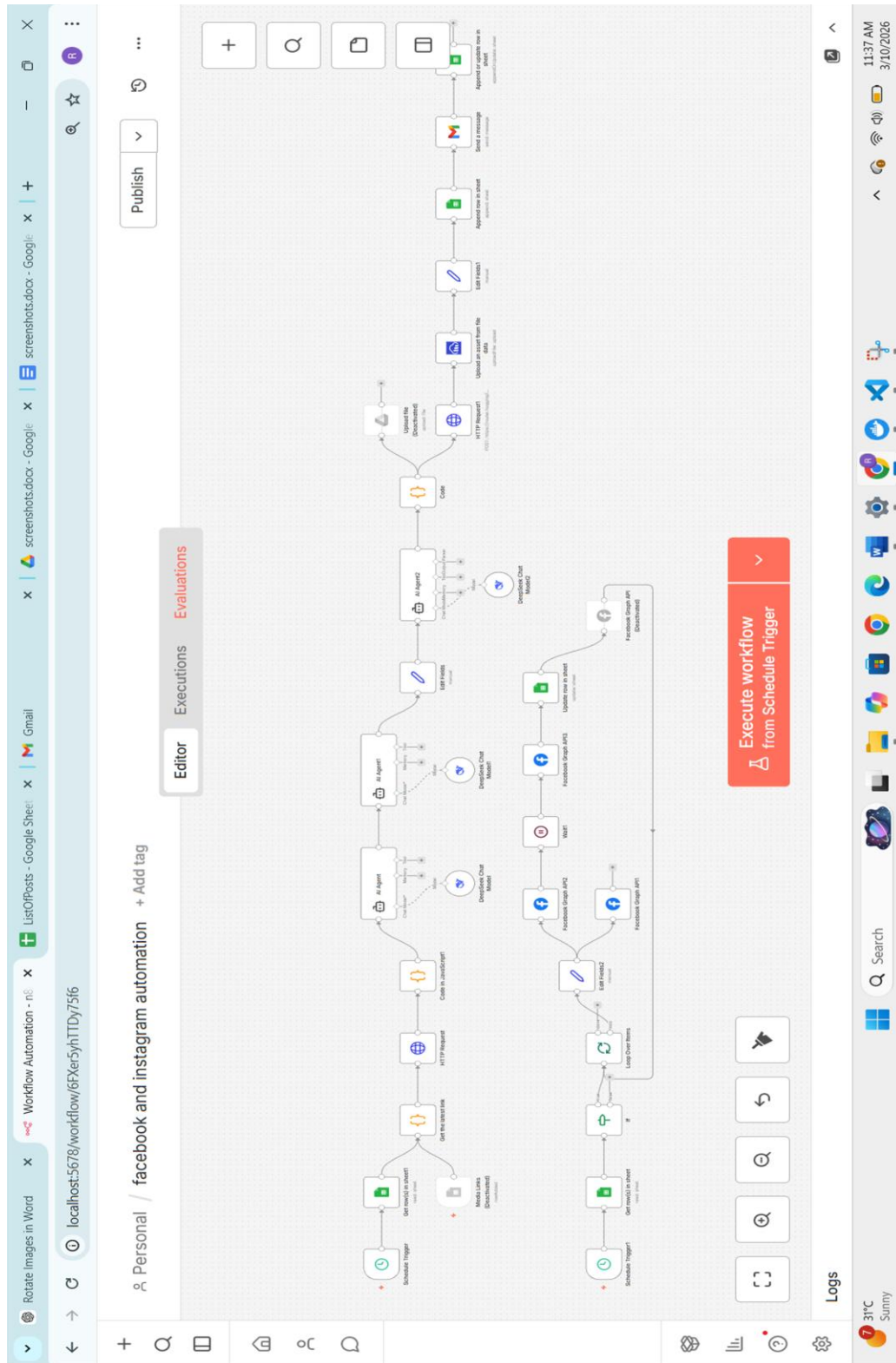


Figure 2.2. 1: Workflow for post generation and posting

Image generation was performed using Google Gemini and HuggingFace models, which produced visuals related to the generated summaries. These images enhanced the visual quality of the posts and improved their potential engagement. The generated images were stored using Cloudinary, providing reliable cloud-based hosting and easy access for publishing.

An approval mechanism was incorporated into the workflow by storing the posts in Google Sheets and sending a mail to the user to allow them to review generated posts before publishing as shown in figure 2.2.2. Once approved, the system automatically posted the content to platforms such as Instagram and Facebook. Overall, the workflow improved efficiency, reduced the time required for manual posting, and demonstrated the effective integration of AI models with automation tools for social media content management.

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File Edit View Insert Format Data Tools Extensions Help							
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AI							
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1	id	blog_url	image_url	draft_content	final_content	status	processed
2	110	https://techyspot.com	empty	China's Vision 2030 aims to make the country a global leader in tech by 2030. The US Customs and Border Protection (CBP) is under fire after a 26-year-old Black man shot a member of the agency's BORTAC unit, a paramilitary unit. The agent was a member of the agency's BORTAC unit. The CBP's reliance on paramilitary units has been a topic of controversy for years. The agency has continued to expand its use of these units, from 100 agents to 1,000 agents. The agency has continued to expand its use of these units, from 100 agents to 1,000 agents. The agency has continued to expand its use of these units, from 100 agents to 1,000 agents.	China's Vision 2030 aims to make the country a global leader in tech by 2030. The US Customs and Border Protection (CBP) is under fire after a 26-year-old Black man shot a member of the agency's BORTAC unit, a paramilitary unit. The agent was a member of the agency's BORTAC unit. The CBP's reliance on paramilitary units has been a topic of controversy for years. The agency has continued to expand its use of these units, from 100 agents to 1,000 agents. The agency has continued to expand its use of these units, from 100 agents to 1,000 agents. The agency has continued to expand its use of these units, from 100 agents to 1,000 agents.	RELE...	TRUE
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7							
8							
9							
10							
Convert to table							
Sheet1							
32°C Sunny							
12:15 PM 3/10/2026							

Figure 2.2. 2: Google sheet for storing generated post

2.2.4 OBJECTIVES ACHIEVED

- Implemented workflow automation using n8n
- Automated generation of social media posts
- Integrated AI image generation models
- Implemented automated publishing to social media platforms

2.3 WORK 3 - AI WORKFLOWS INTEGRATION WITH DJANGO APPLICATION

2.3.1 METHODOLOGY

The implementation of AI workflow integration within the Django application involved combining web development, automation workflows and artificial intelligence services. The process began with analyzing the architecture of the existing Django application to identify areas where AI-powered automation could be integrated effectively. Automation workflows were then designed using the n8n platform to perform various tasks based on user inputs received through the web interface.

Several features were then implemented, including generating social media plans, automating post generation, downloading videos from YouTube using a provided URL and extracting visual features from images. These extracted features were stored and later used as templates to generate images during the auto-posting process.

Finally, automation workflows were configured to handle the process of publishing approved posts directly to social media platforms such as Instagram and Facebook. The system ensured that once a social media plan was generated and approved, the corresponding content could be automatically posted without additional manual steps.

2.3.2 DESIGN AND DEVELOPMENT

Key Technologies and Tools Used:

- **Django** – Used as the main web framework to develop the backend system and manage application logic.

- **Python** – Used for implementing backend functionalities, automation scripts, and AI integrations.
- **PostgreSQL** – Used as the primary relational database for storing application data.
- **pgvector** – Used to support vector storage capabilities within PostgreSQL.
- **n8n** – Used for creating automation workflows that interact with the Django application.

Key Features Implemented

- **AI Workflow Integration with Django**

The Django application was connected with automation workflows, enabling the system to trigger AI-based processes directly from the web interface. Users could interact with different AI-powered functionalities through dedicated pages in the application.

- **Social Media Plan Generation**

A feature was developed that allows users to generate structured content plans. It provides an example post meant for day 1 as output. These plans could be reviewed and approved before initiating automated posting tasks.

- **Autoposting to Social Media Platforms**

After a social media plan was generated and approved, the system automatically published posts to platforms such as Instagram and Facebook through automation workflows.

- **YouTube Video Download Feature**

A dedicated page was developed where users can provide a YouTube video URL and download the video directly through the application. This feature demonstrates integration with external content sources.

- **Image Feature Extraction for Template Creation**

A system was implemented where users can upload an image of a social media post. The application extracts key visual characteristics from the image and stores them as a template.

- Template-Based Image Generation

The stored templates can later be used during the autoposting process to generate images that follow a similar visual structure, ensuring consistency in social media posts.

2.3.3 KEY FINDINGS AND RESULTS

The integration of AI workflows with the Django web application demonstrated how automation can significantly simplify content generation and social media management tasks. By connecting the web interface with automation workflows, users were able to generate social media plans and manage content-related operations directly within the system. Figure 2.3.1. shows the plan generation page.

The screenshot displays the 'AutoPost Setup' interface, which is an AI-Driven Content Plan Generator. The page has a light blue header with the 'AI Social' logo on the left and navigation links for 'Home', 'Features', 'Pricing', and 'Tools' in the center. On the right side of the header, there are links for 'Login' and a blue 'Get Started' button. The main content area is titled 'AutoPost Setup' with the subtitle 'AI-Driven Content Plan Generator'. Below this, there is a form with the following fields and options:

- Describe your posting goal:** A large text input area.
- Duration:** A dropdown menu currently showing '7 Days'.
- Daily Time:** A time selection field currently showing '10:00 AM'.
- Use Example Template (Optional):** A dropdown menu currently showing 'None'.
- Select Platforms:** Two checkboxes, 'Instagram' (which is checked) and 'Facebook'.
- Generate AI Plan:** A prominent blue button at the bottom of the form.

Figure 2.3. 1: HTML page for content generation based on prompt

Additional functionalities such as YouTube video downloading and image feature extraction further expanded the capabilities of the application. The ability to upload an image and extract its design characteristics allowed the system to store templates that could later be used to generate visually consistent images during the autoposting process. This approach helped maintain a consistent style across generated social media posts. The extracted characteristics are stored in JSON format as shown in figure 2.3.2.

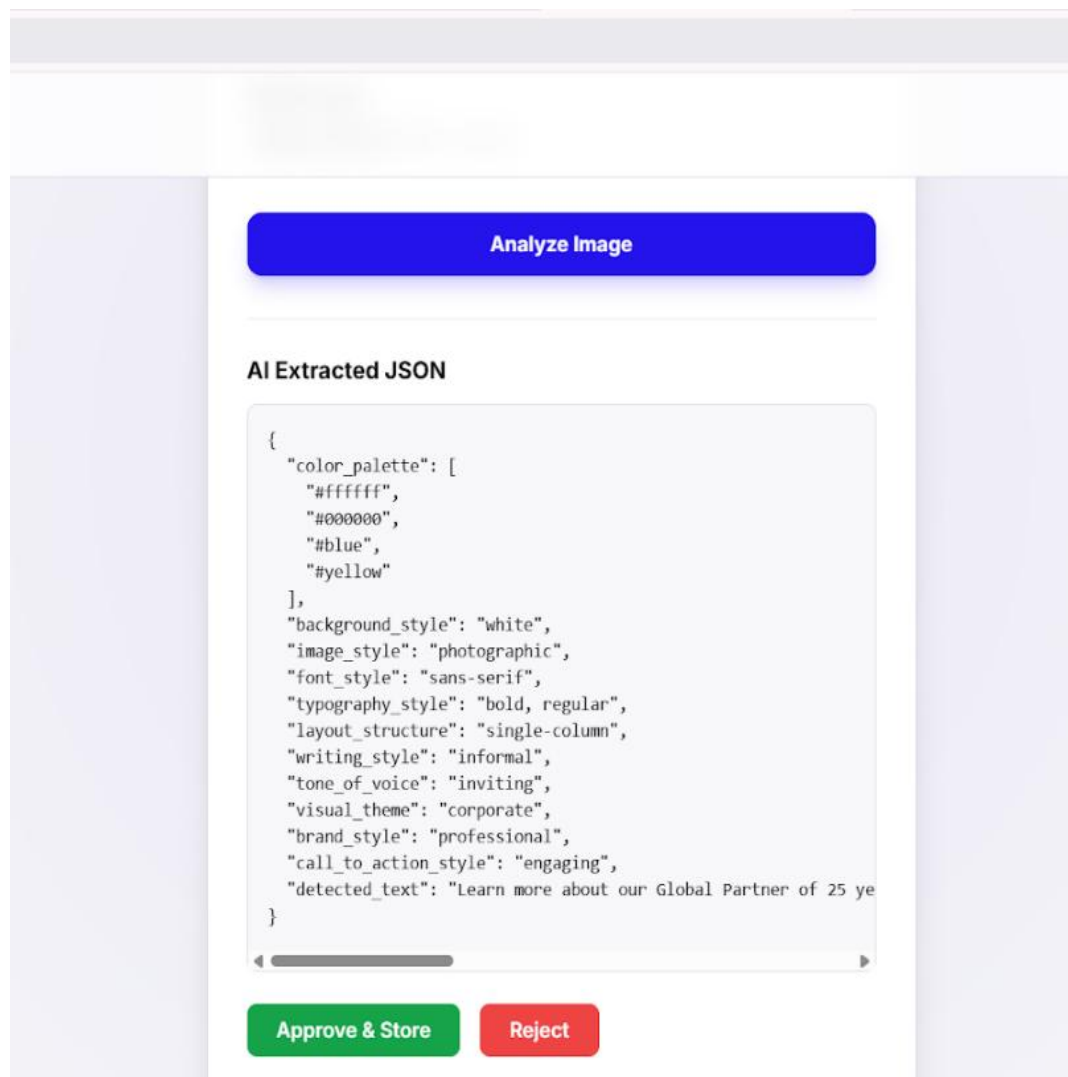


Figure 2.3. 2: JSON output received

The autoposting functionality successfully automated the process of publishing approved content to Instagram and Facebook. Once a social media plan was generated and approved by the user, the workflow automatically handled the creation of posts and their publication on the selected platforms. Figure 2. 4 shows the workflow. Overall, the system demonstrated how AI workflows and web applications can be combined to build intelligent automation solutions for content management and digital marketing tasks.

Figure 2.3.3: Auto-posting workflow for the Django application

OBJECTIVES ACHIEVED

- Successfully integrated AI workflows with a Django-based web application.
- Developed web interfaces that allow users to interact with AI-powered features and automation workflows.
- Implemented a system for generating and managing social media content plans.
- Created functionality to automatically publish approved posts to Instagram and Facebook.
- Developed a feature that allows users to download YouTube videos using a provided URL.
- Implemented image feature extraction from uploaded posts and stored them as reusable templates.
- Enabled template-based image generation to maintain visual consistency during automated social media posting.
- Improved overall efficiency in managing and automating content generation tasks through workflow integration.

CHAPTER 3 – SUMMARY

3.1 CHALLENGES

During the course of the internship, several challenges were encountered while working on different projects and learning new technologies. These challenges played an important role in improving problem-solving abilities and developing a deeper understanding of real-world software development practices.

One of the initial challenges faced during the internship was applying multiple frontend development concepts simultaneously while working on the Medcore Systems demo project. Implementing page layouts, responsive designs, and JavaScript-based functionalities together required careful planning and proper structuring of the code. Achieving proper alignment, responsiveness, and smooth interaction between interface elements initially required several iterations and refinements before the desired behaviour was achieved.

Adapting to changing project requirements was also a significant challenge. During the development process, certain components of the homepage and support pages required modifications, which meant revisiting previously written code and restructuring it to accommodate new functionalities. These changes required careful analysis to ensure that the modifications did not affect the existing features of the application. Handling such changes helped develop adaptability and improved the ability to modify code efficiently without introducing errors.

A major challenge during the internship was learning and working with the n8n workflow automation platform, which was a new technology. Understanding how workflows are structured, how nodes interact with each other, and how different services can be integrated within a workflow required time and experimentation. Designing workflows that could automate tasks such as blog summarization, AI-based image generation, and social media posting involved understanding various triggers, APIs, and data transformation techniques.

Working with containerization technologies such as Docker also presented challenges initially. Configuring containers to run multiple services such as the Django web application, PostgreSQL database, and n8n workflows required

understanding container networking, environment variables, and service dependencies. Properly setting up these components so that they could communicate with each other within the containerized environment required several trials and debugging steps.

Another challenge involved integrating AI services and external APIs, where limitations such as API quotas were encountered. For instance, when the image generation model from Google Gemini reached its usage limits, alternative models had to be researched and integrated into the workflow. Managing multiple responsibilities such as development tasks, meetings, and documentation also required effective time management. Overcoming these challenges through experimentation and guidance significantly improved technical knowledge, problem-solving ability, and confidence in handling real-world development tasks.

3.2 SCOPE FOR FUTURE ENHANCEMENTS

Although the developed system successfully implements automation workflows and AI-assisted content generation, several improvements can be introduced in the future to further enhance the efficiency of the application. One potential enhancement is the integration of additional social media platforms within the automation workflow. Currently, the system supports automated posting to selected platforms such as Instagram and Facebook. Expanding this capability to include platforms like LinkedIn, X and other social media channels would make the system more versatile and useful for organizations that manage multiple online platforms.

The Django application can further be improved by incorporating analytical features that provide insights such as posting frequency, engagement levels and campaign performance. Such analytics dashboards would help users monitor the effectiveness of automated posts and make informed decisions regarding content strategies. Another important enhancement would be the implementation of post security and content moderation mechanisms using Vision-Language Models, which can analyze generated images and text to detect inappropriate or policy-violating content before it is published.

Additional improvements could include enhancing the user interface of the Django application to make it more intuitive and easier to manage. Better visualization of workflow status and simplified approval mechanisms would improve usability. Furthermore, optimizing workflow execution within the automation platform could reduce processing time and resource usage, resulting in improved system performance and operational efficiency.

3.3 CONCLUSION

The internship provided valuable practical exposure to modern software development practices and technologies. During the internship period, work was carried out on multiple development and automation tasks, including frontend interface development and workflow automation using the n8n platform. Automation pipelines were designed to generate summaries from blog content, create AI-generated images, and publish posts to social media platforms. The internship also involved integrating these workflows with a Django-based web application, implementing feature retrieval mechanisms, and configuring containerized environments using Docker to support the deployment of multiple services.

Through these activities, significant technical knowledge was gained in areas such as workflow automation, AI model integration and containerization. Working with technologies such as n8n, Django, PostgreSQL, vector databases and cloud services provided a deeper understanding of how modern applications are structured and deployed. The process of building and optimizing workflows, integrating APIs, and troubleshooting technical issues helped strengthen problem-solving skills and the ability to work with unfamiliar tools and frameworks.

In addition to technical development tasks, the internship also involved research and preparation of technical proposals based on project requirements and discussions with team members. These activities helped improve the ability to analyze requirements, study potential solutions, and document ideas in a structured manner. Overall, the internship experience helped bridge the gap between theoretical knowledge and practical industry practices while contributing significantly to the

development of both technical competencies and professional skills.

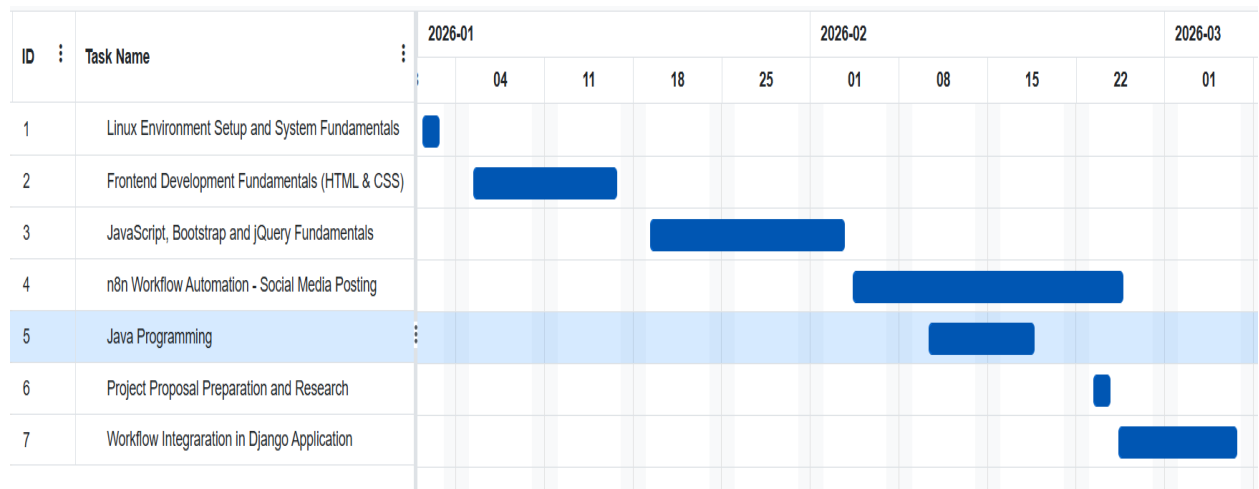


Figure 3.3. 1: Gantt chart showing the internship timeline

3.4 BIBLIOGRAPHY

- Linux Documentation : <https://www.linuxdoc.org/>
- Ubuntu Linux Documentation : <https://help.ubuntu.com>
- Git SCM. Git Documentation : <https://git-scm.com/docs>
- HTML, CSS, JavaScript Tutorial : <https://www.w3schools.com/>
- N8n Documentation : <https://docs.n8n.io/>

ANNEXURE 1 – WEEKLY OVERVIEW OF INTERNSHIP ACTIVITIES

	DAY	DATE	ACTIVITY
WEEK-1	1	01.01.26	• Attended onboarding session and orientation for the internship program which provided an overview of the company, work environment and internship objectives. • Received study materials related to Linux fundamentals • Installed Windows Subsystem for Linux (WSL) and verified Linux environment setup. • Reviewed the fundamentals of the Linux operating system, directory structure and different types of files in Linux.
	2	02.01.26	• Practiced essential Linux directory commands and Linux file commands using the terminal. • Reviewed Linux filesystem hierarchy, basic Unix tools and utilities and Linux I/O redirection concepts. • Attended a meeting with a senior developer regarding mobile application development.
WEEK-2	3	05.01.26	• Participated in a review assessing understanding of Linux concepts. • Reviewed the basics of HTML.
	4	06.01.26	• Practiced different HTML tags and attributes. • Participated in a meeting regarding the review outcomes and agenda for this week. • Studied the fundamentals of CSS.

	DAY	DATE	ACTIVITY
WEEK-2	5	07.01.26	• Practiced topics related to hyperlinks and website navigation. • Covered styling topics including to display properties, layouts and image styling. • Got assigned to create homepage for project titled Medicore Systems based on a given design reference. Started working on the header section of the webpage.
	6	08.01.26	• Worked on designing the middle section of the webpage, which includes the title, description of the system, action buttons, and an image section showing hospital status information. As shown in figure 2.1.1. • Practiced topics related to images, audio, videos, YouTube integration and image maps. • Attended a session on corporate planning, gaining insights into client segmentation, outreach strategies, and long-term business development practices.
	7	09.01.26	• Completed the assigned task and underwent evaluation by a developer. The code implementation is shown in Appendix 1. • Participated in an informative session on NVIDIA, Agentic AI, and related tools.
WEEK-3	8	12.01.26	• Strengthened technical knowledge of HTML5 and CSS3 by implementing links, navigation bars, bookmark links, file links, and iframes.
	9	13.01.26	• Exercised the knowledge of HTML forms by creating a Patient Registration Form for a hospital management system.

	DAY	DATE	ACTIVITY
WEEK-3	10	14.01.26	Strengthened technical knowledge on types of CSS selectors by implementing them by implementation in the project Medicores Systems.
	11	15.01.26	- Participated in a review assessing understanding of frontend technologies. - Practiced responsive web design by implementing CSS media queries to adjust navbar layout and button alignment for different screen sizes in the project Medicores Systems.
	12	16.01.26	- Studied Linux system uses, processes, and job control, and gained hands-on experience with environment variables and the vi editor. - Designed and implemented an automated activity logging system using cron and python to periodically capture system state and file modifications.
WEEK-4	13	19.01.26	- Delivered a technical session on Jenkins, highlighting its architecture, features, and role in continuous integration and deployment. - Attended sessions on Kubernetes, CI/CD pipelines and Docker. - Strengthened knowledge on JavaScript fundamentals and implemented form validation, auto-capitalization of patient names and character count for notes in Patient Registration forms.

	DAY	DATE	ACTIVITY
WEEK-4	14	20.01.26	- Explored advanced JavaScript concepts including functions, objects, and function context methods and related topics. - Implemented a Smart Reminder System which allows users to create time-based reminders, manage reminder information and trigger scheduled alerts.
	15	21.01.26	- Gained practical understanding of JavaScript arrays, data types and class-based object modeling. - Applied these concepts in the Patient Registration form to manage patient information using structured objects, maintain multiple patient records and dynamically display patient details on the interface.
	16	22.01.26	- Continued practicing JavaScript through demo implementations. - Studied JavaScript DOM and JQuery concepts.
	17	23.01.26	JavaScript functionalities were implemented in the demo project Medicore Systems. A form section was added and form validation was performed.
WEEK-5	18	27.01.26	- Removed the form section from the homepage of the demo project Medicore Systems and implemented progress bar animation, hospital bed count updation and alerts. - Studied jQuery concepts. - Participated in a meeting discussing work to be done this week.

	DAY	DATE	ACTIVITY
WEEK-5	19	28.01.26	Studied Bootstrap and created login page for the demo project Medicores Systems. Also performed form validation. Refer to Appendix 2 for code.
	20	29.01.26	Created registration page for Medicores Systems and implemented validation for each input taken. As depicted in figure 2.1.2.
	21	30.01.26	- Continued practicing bootstrap concepts. - Participated in a review assessing understanding of JavaScript and the work completed in the demo project.
WEEK-6	22	02.02.26	Started work on a Support page on the demo project Medicores Systems. Created the FAQ section.
	23	03.02.26	- Revised jQuery concepts and completed the Support page on the demo project. Added a contact support form. - Added JavaScript functionality for FAQ toggle and form validation. Figure 2.1.3 shows the Support page created.
	24	04.02.26	- Got assigned to an automation project which is to be done using the n8n platform. - Started learning the basics of n8n and about workflow automation. - Practiced on implementing blog summarization using the Groq chat model.
	25	05.02.26	- Created required social media accounts required to work on the project. - Worked on posting a static image and caption on Instagram using n8n on button click. - Attended a meeting discussing the project and about work to be done in the coming weeks.

	DAY	DATE	ACTIVITY
WEEK-6	26	06.02.26	- Integrated the summarization and posting workflows. The complete workflow is shown in figure 2.2.1. - Implemented an approval section for the posts so that the user can verify the content before they are posted. As shown in figure 2.2.2.
WEEK-7	27	09.02.26	- Worked on the n8n project after receiving Google Gemini credentials. Generated image each time a post is created out of the input blog URL using imagen 4.0 fast model. - Parallelized the posting and content creation modules so as to ensure same post is not being posted twice. - Started reviewing Java fundamentals including concepts like compilation process and basic syntax.
	28	10.02.26	- Attended a meeting with a client requiring a web application to sell skincare products. Discussed about client situation, possible timeline for product development and other details. - Attended an introductory session on Java and SpringBoot to gain foundational knowledge of backend development. - Strengthened technical knowledge of Java object-oriented concepts and practiced the given question (Bank Management App).
	29	11.02.26	- Prepared a questionnaire to collect client requirements and discussed it with mentor. - Conducted a session on n8n automation with the goal of explaining the platform and and discussed the Instagram posting workflow created. - Attended a meeting discussing the project and about work to be done in the coming weeks.

			Continued practicing Java concepts like encapsulation and inheritance and modified the Bank Management app as specified in the question.
--	--	--	--

	DAY	DATE	ACTIVITY
WEEK-7	30	12.02.26	- Replaced the manual trigger of the workflow with a schedule trigger to run the workflow at regular intervals. Separated the production into 2 workflows due to their schedule times being different. - Continued practiced object-oriented concepts like polymorphism in Java by creating an interface AccountOperations for the Bank Management application and modifying the program as asked.
WEEK-8	31	16.02.26	- Created and configured a Cloudinary account to utilize as cloud storage for AI-generated images and to generate secure public URLs. - Implemented functionality to publish AI-generated images to Instagram. - Practiced Java concepts like abstract classes and interfaces and completed the given task.
	32	17.02.26	- Strengthened knowledge of collections in Java focusing on List and Maps. Created a List Account and implemented adding, deleting and displaying all accounts in the Bank Management App. - Practiced custom exceptions in Java. Integrated the concept in the Bank Management App by creating a custom exception which gets thrown during money withdrawal. Installed and configured Docker and migrated the majority of application workflows to n8n container to enable local hosting.

	DAY	DATE	ACTIVITY
WEEK-8	33	18.02.26	- Replaced the Google Gemini imagen model with Hugging Face shnell model due to API quota limitations. - Completed the updated workflow and submitted the demonstration to the concerned authority. - Studied about Java streams and lambda expressions.
	34	19.02.26	Implemented automated posting of generated content and AI-generated images to a Facebook page and integrated it with the existing workflow pipeline.
	35	20.02.26	- Created an app in meta developers account and practiced sending the content generated as a WhatsApp message using a test number. - Participated in a team building activity to enhance collaboration and communication skills.
WEEK-9	36	23.02.26	- Started preparing the initial project proposal based on the requirements and guidance provided by the mentor. - Conducted research on the client organization and their services to gain relevant background knowledge. Received and reviewed the company database credentials to better understand the organizational structure. - Prepared and submitted the initial draft of the proposal for review.

	DAY	DATE	ACTIVITY
WEEK-9	37	24.02.26	- Received feedback on the initial proposal and was advised to explore solutions focused on specific sub-regions. - Researched relevant tools, including Google Search Console and Google Analytics, to support this requirement. - Incorporated the findings into the existing solution and submitted the revised draft for review.
	38	25.02.26	- Attended a meeting to discuss the project's progress and upcoming tasks. - Set up a Docker container incorporating the Django web application, PostgreSQL, and n8n services. Received instructions to focus on workflow integration and the implementation of RAG and began initial work on these components.
	39	26.02.26	- Practiced implementing webhooks by developing a webpage that accepts a URL as input and generates a summary of its content as output. - Applied this knowledge by integrating an n8n workflow into the project, creating a webpage that accepts a prompt as input and generates an image as output. The code implementation is shown in Appendix 3.

	DAY	DATE	ACTIVITY
WEEK-9	40	27.02.26	- Developed a RAG agent in n8n using the Supabase vector store to enable accurate query responses based on the course-related data it was provided. Configured the workflow to retrieve relevant information from the vector database and generate context-aware responses. - Installed and configured the pgvector extension within the Docker container to support vector storage and similarity search, ensuring proper integration with the PostgreSQL database and enabling efficient retrieval functionality for the agent. Participated in a team building activities to enhance collaboration and communication skills.
WEEK-10	41	02.03.26	- Participated in a discussion to understand the requirements of a new client and contributed to brainstorming potential solution approaches. Based on the discussion, prepared the initial draft of the project proposal and submitted it for review. - Continued working on the RAG implementation on the Django project. Created an embeddings table in the database to store campaign ideas as vector embeddings, enabling efficient semantic retrieval for use within the RAG workflow.
	42	03.03.26	- Continued working on the RAG implementation in the Django project. - Implemented a feature in the Django project to generate and manage social media post plans, including an approval mechanism. Conducted testing across multiple scenarios to ensure the functionality worked reliably and produced consistent results. As depicted in figure 2.3.1.

	DAY	DATE	ACTIVITY
WEEK-10	43	04.03.26	Completed the implementation of the social media posting automation workflow within the Django application. Published the workflow to enable automated execution and verified that it functioned correctly and met the intended requirements.
	44	05.03.26	- Optimized the workflow in the Django application to reduce the use of AI agents. - Implemented functionality to download a YouTube video based on a provided URL. Developed a corresponding HTML page to allow users to input the video link and initiate the download process through the application. Refer to Appendix 4.
	45	06.03.26	- Implemented functionality to extract relevant information from sample posts using a VLM model and store it in the database. Developed an HTML page to facilitate the input and processing of this data. The extracted output is shown in figure 2.3.2. - Modified the social media posting automation workflow to generate post images based on a predefined template, ensuring greater consistency in the generated visual content. As shown in figure 2.3.3. - Participated in the organization's International Women's Day celebrations and related activities.

APPENDIX

APPENDIX 1: HTML CODE OF HOMEPAGE OF MEDICORE PROJECTS

```
<!DOCTYPE html>
```

```
<html>
```

```

<head>
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Medicore Systems</title>
  <link rel="stylesheet" type="text/css" href="styles.css">
  <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/7.0.1/css/all.min.css" integrity="sha512-
2SwdPD6INvrV/lHTZbO2nodKhrnDdJK9/kg2XD1r9uGqPo1cUbujc+IYdlYdEErWNu
69gVcYgdxlmVmzTWnetw==" crossorigin="anonymous" referrerpolicy="no-
referrer" />
</head>
<body>
  <header>
    <div class="top">
      <i class="fa-regular fa-square-plus" style="margin-left: 50px; margin-right: 5px;
border: 2px hidden rgb(231, 219, 244); border-radius: 4px; background-color:
rgb(222, 225, 241); padding: 9px 3px;"> </i>
      <b style="font-size: 200%; margin-top: auto;"> Medicore Systems</b>
    </div>
    <ul>
      <li><a href="#features">Features</a></li>
      <li><a href="#solutions">Solutions</a></li>
      <li><a href="#testimonials">Testimonials</a></li>
      <li><a href="#support">Support</a></li>
      <li><a href="login.html" style="background-color: lightgrey; border-radius:
10px;"><b>Log In</b></a></li>
      <li><a href="#requestdemo" style="background-color: rgb(66, 98, 225);
border-radius: 10px; color: white;">Request Demo</a></li>
    </ul>
  </header>

```

```
<div id="middle">
  <!-- left-part -->
  <div style="width:50%;">
    <div class="bluebullet"> <b>🕒 v2.4 Now Available</b></div>
    <h1 style="font-size:250%; margin-bottom: 0px;">Streamline Patient<br>Care
with <br>Intelligent <br>Management</h1><br>
    <p style="margin-top: 0px; color: rgb(83, 81, 81); font-size:125%;">The all-in-one
platform for scheduling, electronic health<br>
records, and resource allocation. Built for modern<br>
healthcare providers who value efficiency.</p>
    <div class="buttons">

      <button id="btn1" style="background-color: rgb(66, 98, 225); color: white;">Start
Free Trial</button>

      <button id="btn2" style="background-color: white;">View Features</button>
    </div>
    <div style="padding-top: 14px; color: rgb(83, 81, 81); font-size: small;">
      <i class="fa-regular fa-circle-check" style="color: #20e741;"></i> HIPAA
Compliant
      <i class="fa-regular fa-circle-check" style="color: #20e741;padding-left:
5px;"></i> No Credit Card
    </div>
  </div>
  <!-- right image -->
  <div style="width:40%;">
    <br><br>
    
    <div style="font-size: 12px; background-color: white; border-radius: 8px; height:
auto; width: 75%;position: relative; top: -100px; left: 20px; padding: 5px;">
```

```

<b>Emergency Room Capacity </b>
<b style="border-radius: 6px; font-size: 12px; border-style: hidden; background-
color: #20e741; color:darkgreen; height: 5px; width: 45px; padding:
2px;">Optimal</b><br>
<progress id="prg" value="80" max="100" style="width:
90%;"></progress><br>
<span id="bedsText"style="font-size: x-small; color: rgb(83, 81, 81);">24 beds
available: Updated 2 mins ago</span>
</div>
</div>
</div>
<footer style="padding: 20px 0px 20px 0px;">
<p style="text-align: center; font-weight: bold; color: rgba(69, 69, 69,
0.986)">TRUSTED BY LEADING HEALTHCARE FACILITIES</p><br>
<div class="foot">
<div class="ficons" style="margin-right: 50px;"><i class="fa-solid fa-star-of-
life"></i> CityGeneral</div>
<div class="ficons"style="margin-right: 50px;"><i class="fa-sharp fa-solid fa-
heart-pulse"></i> HealthCare</div>
<div class="ficons"style="margin-right: 50px;"><i class="fa-solid fa-prescription-
bottle"></i> KinderMed</div>
<div class="ficons"><i class="fa-solid fa-brain"></i> NeuroLink</div>
<!-- 191 191 191 -->
</div>
</footer>
<script>
const progress = document.getElementById("prg");
let value = 0;
const target = 80;

```

```
.....
const interval = setInterval(() => {
  if (value >= target) {
    clearInterval(interval);
  } else {
    value++;
    progress.value = value;
  }
}, 15);

// alerts for buttons
const btn1 = document.getElementById("btn1");
btn1.onclick = function() {
  if (btn1.innerText === "Start Free Trial") {
    alert("Free Trial Started!");
    btn1.innerText = "Stop Free Trial";
  }
  else {
    const confirmStop = confirm("Are you sure you want to stop the free trial?");

    if (confirmStop) {
      alert("Free Trial Stopped!");
      btn1.innerText = "Start Free Trial";
    }
  }
};

document.getElementById("btn2").onclick = function() {
  alert("Features Page Coming Soon!");
};

setInterval(function() {
  const beds = 22 + Math.floor(Math.random() * 5);
  .....

```

```

.....
document.getElementById("bedsText").innerText =
    beds + " beds available • Updated just now";
}, 3000);
</script>
</body>
</html>

```

APPENDIX 2: HTML CODE FOR LOGIN PAGE OF MEDICORE SYSTEMS

```

<!DOCTYPE html>
<html>
<head>
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Login</title>

    <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"
rel="stylesheet">
    <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/7.0.1/css/all.min.css">
</head>
<body>
<div class="container">
    <div class="login-card">
        <!-- Logo / Title -->
        <div class="text-center mb-3">
            <div class="logo">
                Medcore Systems
            </div>
            <p class="small-text">Secure login for hospital staff</p>
        </div>

```

```
.....
<!-- Login Form -->
<form>
  <div class="mb-3">
    <label class="form-label fw-bold">Username</label>
    <input type="text" id="username" class="form-control" placeholder="Enter
your email or username">
  </div>

  <div class="mb-3">
    <label class="form-label fw-bold">Password</label>
    <input type="password" id="password" class="form-control"
placeholder="Enter your password">
  </div>

  <div class="mb-3">
    <label class="form-label fw-bold">Role</label>
    <select id="role" class="form-select">
      <option>Doctor</option>
      <option>Nurse</option>
      <option>Admin</option>
      <option>Staff</option>
    </select>
  </div>

  <div class="d-flex justify-content-between align-items-center mb-3">
    <div class="form-check">
      <input class="form-check-input" type="checkbox">
      <label class="form-check-label small-text">Remember me</label>
    </div>
    <a href="#" class="small-text">Forgot password?</a>
  </div>
.....
```

```
.....
</div>
  <button type="button" class="btn btn-primary text-light fw-bold w-100"
id="loginBtn">
    Log In
  </button>
  <div class="text-center mt-3 small-text">
    New user? <a href="register.html">Register here</a>
  </div>
</form>

  <div class="text-center mt-3 small-text">
<a href="indexv2.html">Back to Home</a>
  </div>
</div>
</div>
<script>
document.getElementById("loginBtn").onclick = function() {
  const username = document.getElementById("username").value;
  const password = document.getElementById("password").value;
  const role = document.getElementById("role").value;
  if (username === "" || password === "" || role === "") {
    alert("Please fill all fields!");
  }
  else {
    alert("Login feature coming soon!");
  }
};
</script>
</body>
</html>
.....
```

APPENDIX 3: JAVASCRIPT CODE FOR IMAGE GENERATION IN THE DJANGO APPLICATION

```
<script>
let generatedBlob = null; // Store blob for download
document.getElementById("contentForm").addEventListener("submit", async
function(e) {
    e.preventDefault();
    const promptValue = document.getElementById("promptInput").value;
    const generateImageValue =
document.getElementById("generateImage").checked;
    const status = document.getElementById("status");
    const loading = document.getElementById("loading");
    const resultDiv = document.getElementById("result");
    const output = document.getElementById("output");
    // Reset UI
    status.innerText = "Processing...";
    resultDiv.classList.add("hidden");
    output.innerHTML = ""; // Clear previous content
    loading.classList.remove("hidden");
    try {
        const response = await fetch("http://localhost:5678/webhook-test/image-test-
form", {
            method: "POST",
            headers: { "Content-Type": "application/json" },
            body: JSON.stringify({
                prompt: promptValue,
                generateImage: generateImageValue
            })
        });
    }
});
```

```
    // Treat response as binary blob
    const blob = await response.blob();
    generatedBlob = blob; // store for download
    loading.classList.add("hidden");
    status.innerText = "";
    resultDiv.classList.remove("hidden");
    // Create image element
    const img = document.createElement("img");
    img.src = URL.createObjectURL(blob);
    img.alt = "Generated Image";
    img.className = "rounded-xl w-full max-w-md mx-auto";
    output.appendChild(img);

} catch (error) {
    loading.classList.add("hidden");
    status.innerText = "Error: " + error.message;
}
});

// Download PNG
document.getElementById("downloadPng").addEventListener("click", function() {
    if (!generatedBlob) return;

    const link = document.createElement("a");
    link.href = URL.createObjectURL(generatedBlob);
    link.download = "generated-image.png";
    link.click();
});
</script>
```

APPENDIX 4: FUNCTION DEFINED IN VIEWS.PY FOR YOUTUBE VIDEO DOWNLOAD

```
def youtube_download(request):
    if request.method == "POST":
        url = request.POST.get("url")
        format_id = request.POST.get("format_id")
        try:
            ydl_opts = {
                "format": format_id,
                "outtmpl": f'{DOWNLOAD_PATH}/{%(title)s.%(ext)s}',
                "quiet": True
            }
            with yt_dlp.YoutubeDL(ydl_opts) as ydl:
                info = ydl.extract_info(url, download=True)
                messages.success(request, f"Downloaded: {info.get('title')}")
        except Exception as e:
            messages.error(request, f"Download failed: {str(e)}")
    return redirect("/youtube/")
```

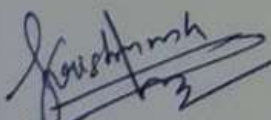
Attendance Certificate

TO WHOM IT MAY CONCERN

This is to certify that **Ms. Riya Mathew** (Aadhaar No: 8242 3423 1112) of Rajagiri College of Social Sciences (Autonomous), Kalamassery, has successfully completed her Internship as a **Python Full-Stack Developer Trainee** at GDK TechySpot LLP, Ernakulam Town, Kerala, from **01 January 2026** to **06 March 2026**.

During the internship period, she maintained 99% attendance and demonstrated good discipline and commitment towards the assigned tasks.

We wish her all the best for her future endeavors.



(Director & Authorized Signatory)

Krishnan G
Chief Executive Officer
GDK Techyspot LLP



GDK TECHYSPOT LLP

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